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Random forests and foundation-model representations

Week 10: Ensembles, feature importance, and the first three-way model comparison.

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By the end of this week you should be able to

Learning outcomes served: LO1, LO2, LO3, LO5

- Explain bagging and why averaging decorrelated trees reduces variance.
- Describe how random feature subsetting at each split further decorrelates the ensemble.
- Compute and interpret out-of-bag error.
- Interpret feature importances correctly, including the correlated-feature and cardinality caveats.
- Compare engineered features, foundation-model embeddings and a fine-tuned model on a shared task, on more than accuracy.

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The biological problem

Part 1 · Why anyone cares about this week's method

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Biological context

- Random forests remain one of the most reliable defaults for tabular biological data and are extensively used in genomics and clinical prediction. They need little tuning, handle mixed feature types, are robust to outliers and irrelevant features, and provide a feature-importance ranking.

- That last property is what makes them attractive to biologists, and also what most often gets over-interpreted.

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The method

Part 2 · Concepts, then the mathematics

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The wisdom of crowds

Averaging many independent, individually mediocre estimates can beat any one of them. The word doing the work is "independent", and the whole design of a random forest is an attempt to buy independence.

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Bagging

Bootstrap aggregating: fit each tree on a resample drawn with replacement, then average. Individual trees are high-variance; the average is not.

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Key concepts

Random feature subsetting At each split, consider only a random subset of features. Without this, every tree would pick the same dominant feature at the root and the ensemble would be highly correlated.

Out-of-bag error Roughly a third of samples are omitted from each bootstrap, giving a free held-out estimate without a separate validation set.

Feature importance, with care Impurity-based importance is biased towards high-cardinality and continuous features, and splits importance arbitrarily between correlated features.

Robustness Forests tolerate irrelevant features, outliers and unscaled data far better than k-NN or linear models, which is much of why they are a good default.

Where forests still overfit They resist overfitting in depth but not in protocol. A forest tuned on the test set is overfitted exactly as badly as anything else.

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Bagging: why averaging helps

$$\mathrm{Var}\left(\frac{1}{B}\sum T(b)\right) = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{B}$$

Averaging B estimators reduces variance, but only the second term shrinks with B. The first term is floored by the correlation between the trees, which is why decorrelating them is the whole design of a random forest.

B number of trees

T(b) the prediction of the b-th tree, b = 1 … B

σ^2 variance of a single tree

ρ pairwise correlation between trees

consequence bootstrapping alone leaves ρ high, because every tree splits on the same dominant feature

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Random feature subsetting

$$m = \lfloor \sqrt{p} \rfloor \qquad \text{considered at each split}$$

At every split, consider only a random subset of the features. This is what buys the independence that averaging needs, and it is the difference between a random forest and bagged trees.

- **p** total number of features
- **m** features sampled per split; √p for classification, p/3 for regression
- **OOB** roughly 37% of samples are omitted from each bootstrap, giving a free held-out estimate

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Measured on WDBC, five-fold cross-validation

The simplest model wins. Standard deviations across folds are one to two points, so several of these are not distinguishable.

Model	Accuracy	F1	ROC-AUC	Interpretability
Majority-class baseline	62.7%	—	—	Total
Decision tree, depth 3	91.7%	0.887	0.925	Readable
Decision tree, unpruned	93.8%	0.917	0.936	In principle
Random forest, 500 trees	95.4%	0.938	0.991	Partial
Gradient boosting, 200	95.4%	0.938	0.992	Partial
Logistic regression	97.4%	0.965	0.995	Full
PCA (10) + logistic regression	97.7%	0.969	0.996	Partial

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Feature importance, and what it cannot bear

This is the single most over-interpreted output in applied biological machine learning.

- Impurity-based importance is biased towards continuous and high-cardinality features
- Correlated features split the credit arbitrarily between them
- On WDBC, 'worst area' and 'worst perimeter' score 0.129 and 0.129 because they measure nearly the same thing
- Permutation importance on held-out data is more trustworthy
- Neither is a causal claim, and neither licenses a therapeutic target
- A forest resists overfitting in depth but not in protocol: one tuned on the test set is overfitted like anything else

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Worked example

Part 3 · Three models, and the columns that are not accuracy

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Three models, and the columns that are not accuracy

- Model A, engineered features into a random forest: 95.4 per cent cross-validated accuracy on WDBC, measured. Trains in seconds on a laptop, no GPU, gives a feature ranking a clinician can read.
- Model B, foundation-model embeddings into a random forest: needs a pretrained model appropriate to the data type. For WDBC there is none, because the input is thirty morphometric numbers rather than a sequence, an image or text.
- This row is not "worse"; it is not applicable, and recognising inapplicability is a skill.
- Model C, a fine-tuned foundation model: only in play where a suitable pretrained model exists. It requires a GPU, careful regularisation on small data, and a much more demanding validation protocol.

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Three models, and the columns that are not accuracy (continued)

- The comparison table on this page therefore has columns for training-data requirement, compute cost, interpretability and validation burden alongside accuracy.
- A model that wins on accuracy and loses on the other four is not obviously the right choice, and for a clinical application it is usually the wrong one.
- Note the honest scoreboard on the real data: plain logistic regression reaches 97.4 per cent, PCA followed by logistic regression 97.7 per cent, and the 500-tree random forest 95.4 per cent. The simplest model in the comparison wins. That happens more often than the literature would suggest.

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Compare on the axes that will decide whether the model gets used, not only on the one that is easiest to report.

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Try this interactively

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Random Forest Visualiser

What weight can a feature-importance ranking actually bear?

`weeks/week10.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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Try this interactively

Model Comparison Lab

What changes when you sort by something other than accuracy?

`weeks/week10.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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The foundation-model connection

Part 4 · Embeddings plus a forest

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Embeddings plus a forest

- Where a suitable pretrained model does exist, embeddings into a random forest is a strong and underrated pipeline.
- The forest is cheap, needs no GPU, gives importances over embedding dimensions (though those dimensions rarely mean anything individually), and can be cross-validated honestly in seconds.
- It is a genuine competitor to fine-tuning, and it should be run before fine-tuning is considered, so that fine-tuning has something to beat.

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Interpretation and practical

Part 5 · What the result means, and what you will do with it

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Your forest ranks gene X as the most important predictor of treatment response. A colleague proposes targeting gene X therapeutically. Explain, in terms a non-specialist would follow, why feature importance does not support that step.

Think like a scientist. Bring your answer to the laboratory.

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Ensembles and the limits of feature importance

This week's laboratory · Fit a strong ensemble, then establish how much its feature importances can bear. · 2 hours

- Fit a single decision tree, bagged trees and a random forest with matched settings. Report cross-validated accuracy for all three and explain the ordering you observe.
- Sweep n_estimators from 1 to 500 and plot out-of-bag error. Where does it plateau? What does that tell you about the marginal value of more trees?
- Extract impurity-based feature importances and rank them. Then compute permutation importance on held-out data. Compare the two rankings and account for any disagreement.
- Find the most correlated pair of features in the data. Drop one and refit. What happened to the importance of the survivor, and what does that tell you about reading importances as biological significance?
- Fit logistic regression on the same data. Compare its accuracy with the forest. Discuss what you would report if the simpler model were within one standard deviation of the complex one.
- Build the full comparison table for this dataset: accuracy, F1, ROC-AUC, training time, whether a GPU is needed, and how interpretable each model is. Recommend one model to a clinical collaborator and justify the recommendation on more than accuracy.

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Self-check

- Why does a random forest sample features at each split as well as bootstrapping the rows?
- A) To speed up training
- B) To decorrelate the trees, since bootstrapping alone leaves them all splitting on the same dominant feature
- C) To handle missing values
- D) To reduce memory use

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Self-check

- Two features have importances of 0.129 and 0.129 and are highly correlated. What follows?
- A) Both are causally important
- B) The forest has split the credit between two near-duplicate measurements; neither figure supports a causal claim
- C) One is redundant and should be dropped
- D) The model is overfitting

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References and further reading

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Questions

Next week: continue to Week 11.

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